



Short Communication

Failure and strength of 2D-C/SiC composite under in-plane shear loading at elevated temperatures

K.F. Yan, C.Y. Zhang*, S.R. Qiao, M. Li, D. Han

National Key Laboratory of Thermostructure Composite Materials, Northwestern Polytechnical University, Xi'an 710072, PR China

ARTICLE INFO

Article history:

Received 3 November 2010

Accepted 23 February 2011

Available online 1 March 2011

ABSTRACT

In-plane shear strength (IPSS) of a two-dimensional carbon fiber reinforced silicon carbide composite (C/SiC) was measured by compression of double-notched specimens (DNS) from room temperature to 1873 K. The result indicates that the compression of DNS is an effective method to measure IPSS of C/SiC. A significant dependence of IPSS on temperature was found for C/SiC. IPSS increases with increasing temperature up to 1273 K, and then decreases when the temperature was higher. The main failure modes under the shear loading contain matrix cracking, delamination and pullout of fibers and fiber bundles.

© 2011 Elsevier Ltd. All rights reserved.

1. Introduction

Continuous carbon fiber reinforced silicon carbide composite (C/SiC) overcomes the brittleness of monolithic ceramic and possesses outstanding properties of the matrix and the fiber. Therefore, it was considered as a potential refractory material applied in the aerospace field [1,2]. The mechanical properties of the composite are closely related to the constituent properties and the fiber/matrix interface. Moreover, the strength and failure behaviors of C/SiC depend on how the composite is loaded with respect to the principal material directions, due to the anisotropy of its properties. When holes, notches, attachments or off-axis loading appears in the structural component made of C/SiC, the failure caused by shear deformation might be dominant due to the smaller shear strength of C/SiC than its tensile and compressive strength, the shear deformation could take place either within the plane of composite, i.e. in-plane, or within the plane of thickness, i.e. interlaminar. So far, the mechanical properties under the tension, flexural and interlaminar shear loading were investigated extensively [2–7]. Brondsted [8], Carl [9] and Guan [10] etc. studied the in-plane shear strength (IPSS) of C/SiC at room temperature using Iosipescu tests, where a specimen in the form of a rectangular flat strip with two symmetric central V-notches was loaded by a four-point asymmetric fixture at room temperature [8,11]. It is urgent to understand the in-plane shear behaviors of C/SiC at elevated temperatures. Whereas, there is no a reliable testing method for the determination of the IPSS at elevated temperatures. The compression of double-notched specimens (DNS) was specified to determine the interlaminar shear strength of ceramic matrix

composites at elevated temperature by American Society for Testing and Materials (ASTM) standard (C-1425-05) [12]. Previous results confirmed it was also an effective method to determine IPSS of ceramic matrix composite at high temperatures [13,14].

Therefore, the aim of the present work is to study the IPSS of a C/SiC reinforced by plain woven carbon fiber clothes by compressing the DNS from room temperature to 1873 K. Furthermore, the deformation mechanism was investigated.

2. Experimental procedures

2.1. Material

The two-dimensional carbon fiber reinforced silicon carbide composite (2D-C/SiC) was reinforced by a stack of plain-woven cloth, made of T300 polyacrylonitrile (PAN) carbon fiber. The composite was fabricated by chemical vapor infiltration (CVI). The PAN carbon fibers were coated with pyrocarbon (PyC) with thickness of 200 nm before infiltration of SiC matrix. The density, the fiber volume fraction, and residual porosity of the prepared C/SiC were 2.0 g/cm³, 40% and 17%.

2.2. Experimental

IPSS at elevated temperatures was determined by the compression of DNS. Fig. 1 shows a schematic of the fixture for the compression of DNS [15]. The fixture consisted of a slotted body, one loading piston, one rectangular spacer, a trapezoid spacer and a compliant wedge between the specimen and the trapezoid spacer. The wedge was used to guarantee uniform axial compression of the specimen, and to provide lateral support to prevent buckling. Meanwhile the wedge can minimize the mechanical interference

* Corresponding author. Tel./fax: +86 029 88492084.

E-mail address: cyzhang@nwpu.edu.cn (C.Y. Zhang).

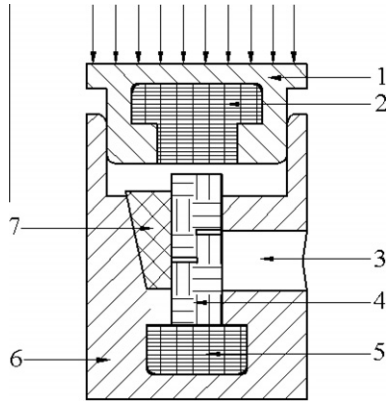


Fig. 1. Schematic of fixture used for the compression of double-notched specimens at elevated temperatures. 1-loading piston, 2-upper cushion, 3-hole for the thermocouple, 4-specimen, 5-lower cushion, 6-slotted body, 7-trapezoid spacer.

between the fixture and the specimen and avoid transversely compressing the specimen at the test temperature. The material used for the manufacture of the fixture was graphite, which was stable and rigid at the test temperature.

The shape and dimensions of DNS is shown in Fig. 2a. In order to verify the validity of IPSS obtained by compression of DNS, the IPSS was also determined by the Iosipescu method [16], which was the adopted as the standard method to measure the IPSS at room temperature. For the Iosipescu method, the geometry of the specimen is shown in Fig. 2b. The measurement was carried out in a mechanical testing machine (Model: CSS-280S-100). The cross-head speed was 0.595 mm/min.

The experiments at room temperature were carried out in air and the experiments at elevated temperatures were performed in a vacuum mechanical testing machine. The applied temperatures were 973 K, 1173 K, 1273 K, 1473 K, 1673 K and 1873 K. The specimens were heated to the test temperature at a rate of 40–50 K/min, followed by a soaking period of 15 min to allow for thermal

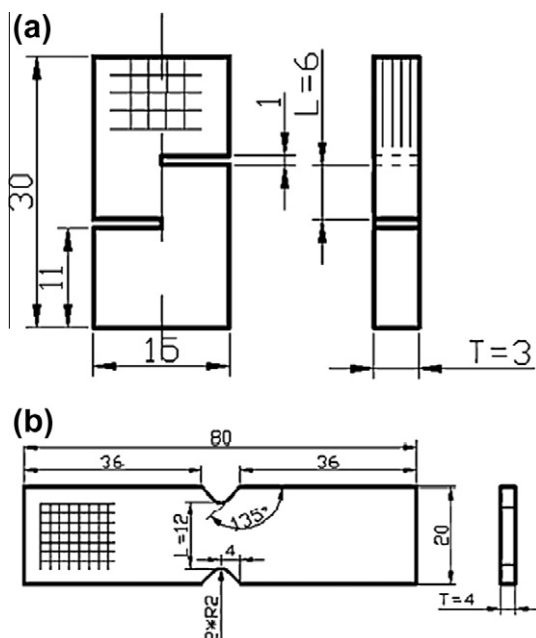


Fig. 2. Geometry of the specimen for: (a) compression of the DNS and (b) Iosipescu method.

equilibrium. A W–Re thermocouple, positioned inside the vacuum chamber and contacting the surface of the specimen directly, was used to monitor and control the temperature. The tests were conducted under a constant displacement rate of 0.595 mm/min. At least seven tests were conducted at each temperature. The IPSS was calculated by the following equation:

$$\text{IPSS} = P_{\max}/LT \quad (1)$$

where $P_{\max}$ is the maximum load, L is the distance between the notches and T is the specimen thickness.

A XTZ-D# continuous stereomicroscope (XTT) was used to reveal the formation and propagation of cracks in the specimen at room temperature. The fracture morphology was observed by a scanning electron microscope (SEM, HITACHI S-4700).

3. Results and discussion

3.1. Comparison of the compression of DNS and the Iosipescu method

Fig. 3 shows the IPSS of C/SiC obtained by the compression of DNS and the Iosipescu method. It can be found that the IPSS of C/SiC measured by the compression of DNS is rather close to that determined by the Iosipescu method. The same trend can also be found for C/C [14]. The failures took place on the plane between two notches, as shown in Fig. 4. It indicates that the failure was caused by the shear loading because the plane is parallel to the loading direction. As a result, the valid IPSS can be obtained by the compression of DNS.

The IPSS of C/SiC is much larger than that of interlaminar shear strength (ILSS) of C/SiC with same reinforcement by about three times regardless of the temperature [7]. The stress distributions in the interlaminar and in-plane shear specimens are same, according to the results of finite element analysis [13]. Therefore, the great difference between the IPSS and ILSS result from the different loading directions.

3.2. In-plane shear strength at elevated temperatures

The IPSS of C/SiC as a function of temperatures is shown in Fig. 5. It can be found that IPSS increases with increasing temperature in the range of 298 K–1273 K, followed by a decrease when the specimens were tested at higher than 1273 K. Moreover, IPSS at elevated temperature is always higher than that at room temperature in the investigated temperature range.

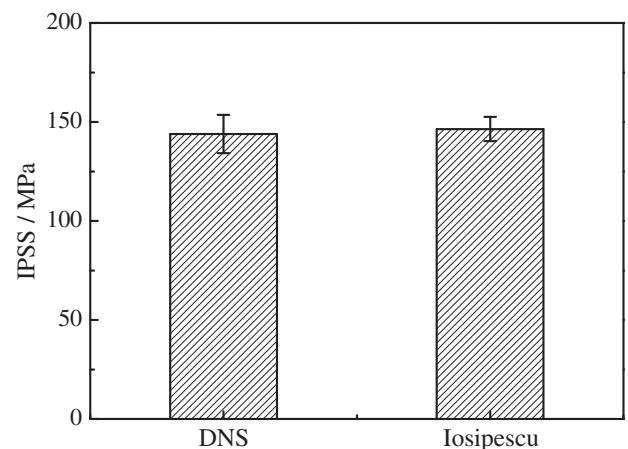


Fig. 3. IPSS of the C/SiC measured by compression of DNS and the Iosipescu method.

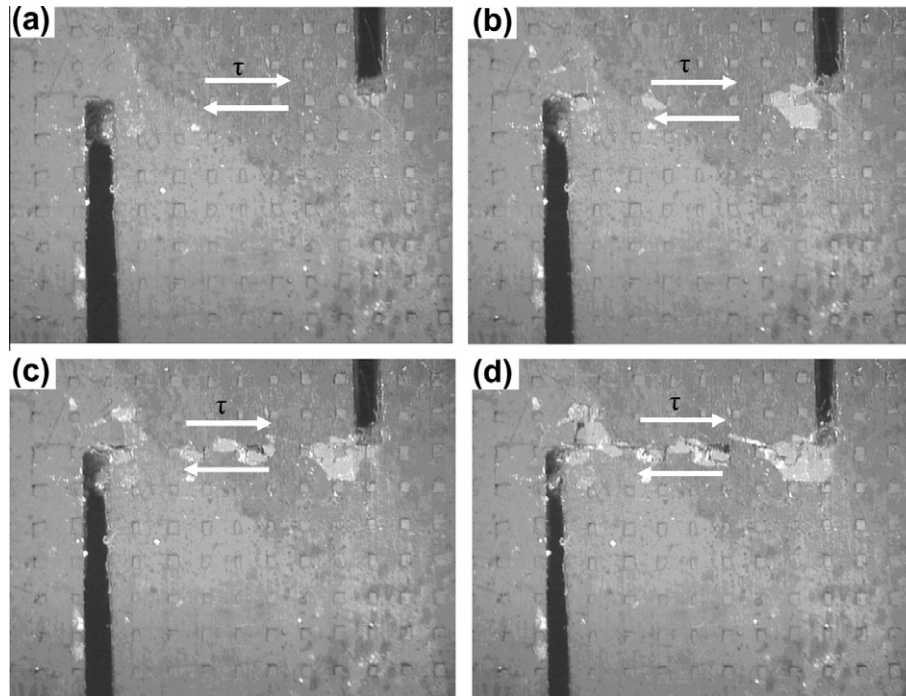


Fig. 4. In situ images of the specimen under in-plane loading at room temperature taken by XTT. (a) 0 s, (b) 130 s, (c) 150 s, (d) 160 s.

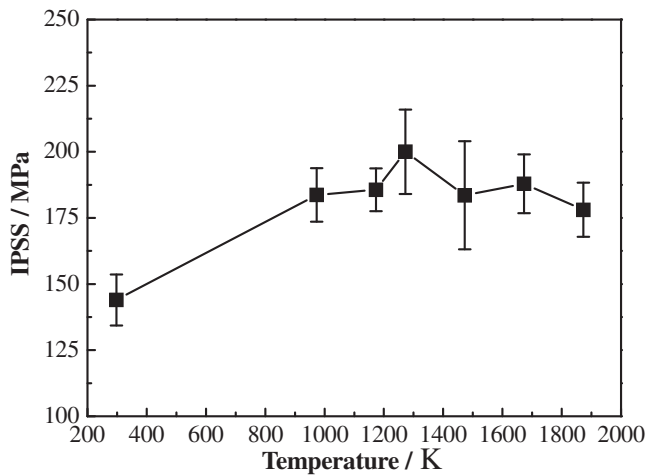


Fig. 5. IPSS of the C/SiC at elevated temperatures. IPSS was measured by compressing DNS at 0.595 mm/min. Besides the experiments at room temperature, those at elevated temperatures were carried out in vacuum.

The variation of the IPSS with temperature is closely related to the thermal residual stress resulted from the mismatch of the coefficient of thermal expansion (CTE). The CTEs of carbon fibers and SiC are $1 \times 10^{-6}/\text{K}$ and $4.8 \times 10^{-6}/\text{K}$, respectively. The mismatch of the CTEs resulted in residual stress in the composite when the materials were cooled from the preparation temperature (1273 K) to room temperature. Though the matrix crack could relax partial stress, tremendous stress still retain in the composites [17]. The axial residual stress in matrix (σ_{rm}) as a function of temperature can be calculated by the following equation [18].

$$\sigma_{rm} = \left(E_m \frac{\phi_2}{\phi_1} \right) \left(\frac{E_f}{E_c} \right) \left(\frac{V_f}{1-\nu} \right) (\alpha_f - \alpha_m) (T_t - T_p) \quad (2)$$

$$\phi_1 = 1 - 0.5 \left(\frac{1-2\nu}{1-\nu} \right) \left(1 - \frac{E_c}{E_f} \right), \phi_2 = 0.5 \left(1 + \frac{E_c}{E_f} \right)$$

where, α_f and α_m are the axial CTEs of fibers and matrix, respectively, T_t the test temperature, T_p the preparation temperature, E_f elastic modulus of PAN carbon fiber, E_c the elastic modulus of C/SiC, ν Poisson's ratio and V_f fiber volume fraction.

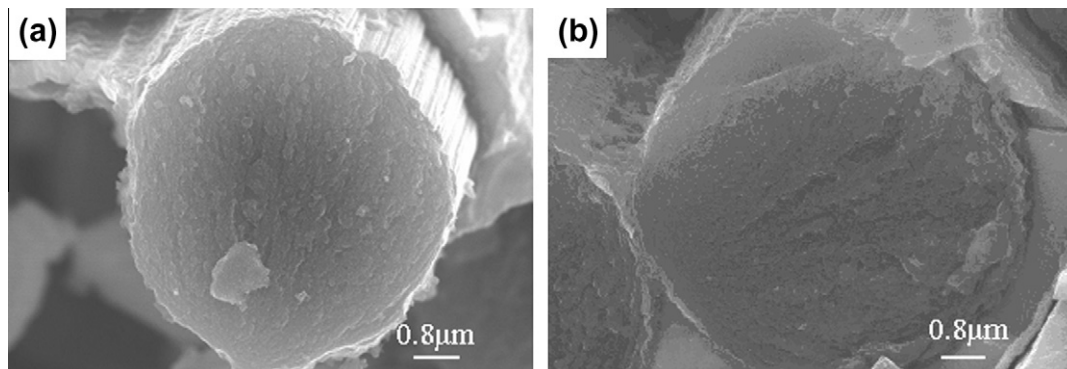


Fig. 6. Fracture surfaces of carbon fiber failed at 1273 K: (a) with and (b) without a mirror of fracture.

The CTE of the fiber is smaller than that of the matrix. According to the above equation, tensile stress present on the fiber/matrix interface at temperature lower than the preparation temperature (1273 K). The tensile interface stress can weaken the bonding strength between the fiber and the matrix. The tensile stress on the interface was gradually released with increase in temperature from room temperature to the preparation temperature, and was negligible at the preparation temperature. The release of the residual tensile stress means that the bonding strength of the fiber/matrix interface was enhanced in the C/SiC at elevated temperatures. The enhancement of the interface bonding gave rise to the increase of the IPSS. The complete relaxation of the residual stress leads to the largest IPSS at 1273 K. The residual stress becomes compressive at above the preparation temperature. On one hand, the compressive stress could enhance the bonding of fiber/matrix. On another hand, much matrix cracks were provoked by the compressive stress and lowered the strength of the SiC matrix. When the weakening produced by the matrix cracking is larger than that of the enhancement of the interface, the IPSS decrease. The trend is the same as the results for the tensile [3], interlaminar [7] shear strength at elevated temperatures.

3.3. Failure mechanisms

Fig. 4 shows four in situ images taken by a XTT microscope. It can be seen that the cracks initiated at the notch roots and the coating around the notch roots broke according to Fig. 4a and b. The cracks propagated towards to the center of the specimen along the plane between two centrally located notches with the increase of the loading, shown in Fig. 4c. Eventually, the cracks aroused from both notches roots met together and resulted in the failure of the composite, shown in Fig. 4d. The reason why the cracks form at the notch roots can be explained by stress concentration around

the notch roots, which had been modeled by the finite element analysis [13].

Moreover, the fracture surfaces of fibers can also provide the information related to the failure process. Once the matrix cracks form, it begins to prorogate. The growth of the crack can be divided into two stages. At the first stage, successive failures of individual fibers took place during the propagation of the matrix cracking. Failure of one fiber leads to the loss of the load carrying ability, so the load carried by the fiber was transferred to the neighboring surviving fibers. The stress redistribution promoted failures of new fibers. At the second stage, the remaining fibers could not carry the applied load and instable fracture took place. The two-stage process can be reflected by the different characteristics of the fracture surfaces of the fibers failed at 1273 K, shown in Fig. 6. The fibers around the notches exhibited a fracture mirror, indicating that failure was initiated at a defect located at the interior of the fibers. These fibers dominated the initial stable stage of the fibers failures (the first stage). Any site of fracture initiation cannot be found for the fibers away from the notches. The corresponding fracture surface of the fibers coincided with the plane of the matrix cracking, indicating that they failed together with the matrix cracking during the second stage.

Fig. 7 shows the morphology of the fracture surfaces at 1273 K. It is found that the different failure modes appear for 90° fibers (parallel to the loading) and 0° fibers (perpendicular to the loading). The 90° fibers and/or fiber bundles fails in form of rupture and pullout, shown in Fig. 7b. However, the cracks prefer to propagate along the interface once the crack met the 0° fibers or fiber bundles due to the smaller bonding strength of the fiber/matrix interface than the ultimate strength of the matrix. Therefore, debonding of fiber/matrix interface always took place, shown in Fig. 7c. Moreover, matrix cracking can also be found on the fracture surfaces (Fig. 7d). So the failure mechanisms of the C/SiC under the in-plane loading include the rupture and pullout of the 90° fibers

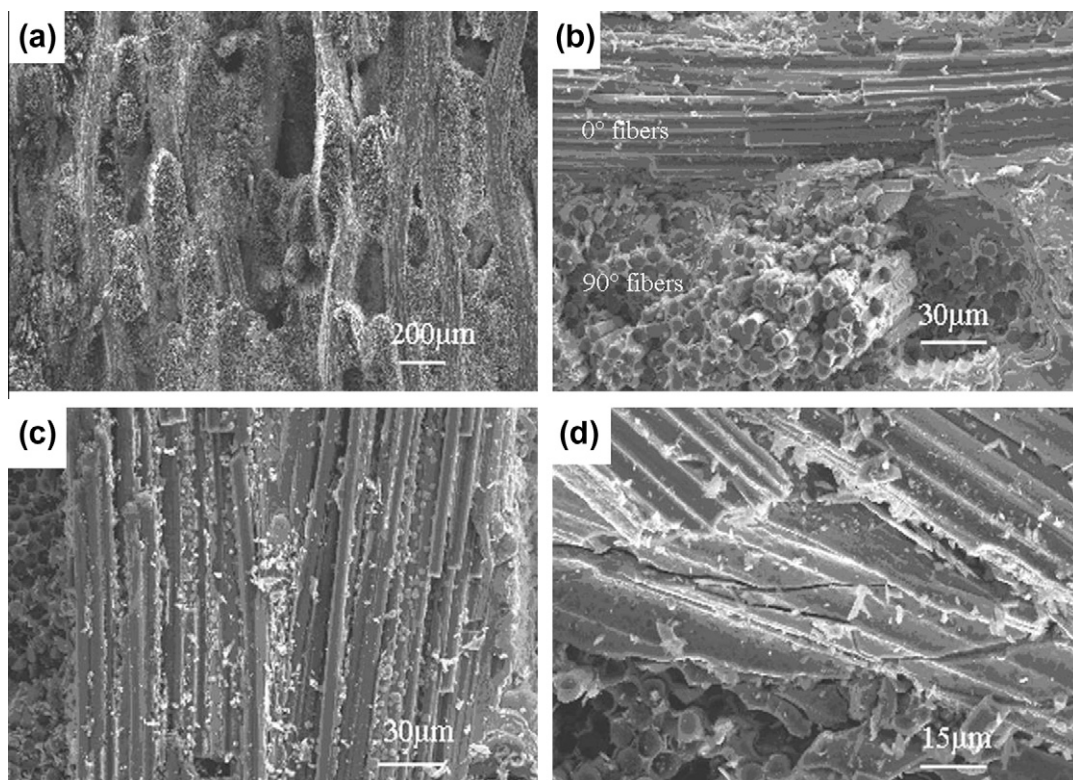


Fig. 7. SEM images on the fracture surface of C/SiC failed at 1273 K.

and/or fiber bundles, debonding of the fiber/matrix interface and matrix cracking.

4. Conclusions

- (1) The compression of DNS is an effective method in measuring the IPSS of C/SiC composite. The obtained value is close to that of the Iosipescu method.
- (2) IPSS of C/SiC composite increases with increasing the temperature until 1273 K, and then decrease at the temperature of higher than 1273 K. IPSS at high temperatures is always higher than that at room temperature in the investigated temperature range.

Acknowledgments

The authors would like to Yongsheng Liu for supplying the samples and Bo Wang for helping us to measure the IPSS at room temperature by the Iosipescu method. Also the authors acknowledge the financial support from National Science Foundation of China (Grant No. 50702045), the Program for New Century Excellent Talents in University (NCET-08-0460) and the 111 Project (B08040).

References

- [1] Evans AG. Perspective on the development of high toughness ceramics. *J Am Ceram Soc* 1990;73:187–206.
- [2] Xu YD, Cheng LF, Zhang LT. Mechanical properties of 3D fiber reinforced C/SiC composites. *Mater Sci Eng A* 2001;300:196–202.
- [3] Qiao SR, Luo GQ, Du SM, Li M. Tensile performance of 3D-C/SiC composites at high temperature. *Mech Sci Technol* 2004;23:335–8.
- [4] Dalmaz A, Reynaud P, Rouby D, Fantozzi G. Mechanical behavior and damage development during cyclic fatigue at high-temperature of a 2.5D carbon/SiC composite. *Compos Sci Technol* 1998;58:693–9.
- [5] Mall S, Engesser JM. Effects of frequency on fatigue behavior of CVI C/SiC at elevated temperature. *Compos Sci Technol* 2006;66:863–74.
- [6] Qiao SR, Li M, Zhang LT. Flexural performance of 3D-C/SiC composites at high temperature and influence of post heat treatment. *J Mech* 2003;25:495–8.
- [7] Wu Q, Zhang CY, Qiao SR. High temperature interlaminar shear strength of two dimensional carbon fiber reinforced (BC_x-SiC)_n composites. *J Chin Ceram Soc* 2009;37:867–70.
- [8] Brondsted P, Heredia FE, Evans AG. In-plane shear properties of 2-D ceramic matrix composites. *J Am Ceram Soc* 1994;77:2569–74.
- [9] Carl C, Heredia FE, Evans AG. In-plane shear properties of several ceramic-matrix composites. *J Am Ceram Soc* 1995;78:2065–78.
- [10] Guang GY, Jiao GQ, Zhang ZG. In-plane shear fracture characteristics of plane in 2D woven C/SiC composite. *Mech Sci Technol* 2005;24:515–7.
- [11] ASTM Designation C-1292-00. Standard test method for shear strength of continuous fiber-reinforced advanced ceramics at ambient temperatures. ASTM Designation; 2005.
- [12] ASTM Designation C-1425-05. Standard test method for interlaminar shear strength of 1-D and 2-D continuous fiber-reinforced advanced ceramics at ambient temperatures. ASTM Designation; 2005.
- [13] Ünal Ö, Bansal NP. In-plane and interlaminar shear strength of a unidirectional Hi-Nicalon fiber-reinforced celsian matrix composite. *Ceram Int* 2002;28(5):527–40.
- [14] Yan KF, Zhang CY, Qiao SR. In-plane shear strength of a carbon/carbon composite at different loading and temperatures. *Mater Sci Eng A* 2011;528(3):1458–62.
- [15] Zhang CY, Qiao SR, Yan KF. Clamp for testing shearing property of continuous fiber reinforced composite material. PRC utility model patent Publication; 2010 [patent number: ZL 2009 2003 3895.3].
- [16] ASTM Designation C-1292-00. Standard test method for shear strength of continuous fiber-reinforced advanced ceramics at ambient temperatures. ASTM Designation; 2005.
- [17] Jiang YQ, Li G, Wang WB. The analysis and measurement of residual stress in ceramic matrix composites interface. In: Tenth international conference on composite materials. IV. Characterization and ceramic matrix composites. British Columbia; Canada: Whistler; 14–18 August, 1995. p. 665–72.
- [18] Budiansky B, Hutchinson JW, Evans AG. Matrix fracture in fiber reinforced ceramics. *J Mech Phys Solids* 1986;34(2):167–89.